Math 121: Commutative Algebra

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Worksheet 2.3: Valuation Rings

Fix a field $\mathbb{K}$. In this worksheet we will explore a special, but important type of rings called valuation rings.

Definition 0.1. *A* valuation ring *of a field* $\mathbb{K}$ *is a subring* $R \subset \mathbb{K}$ *with the property that for all* $x \in \mathbb{K}^\times$ *either* $x \in R$ *or* $x^{-1} \in R$*.*

Note that, since a valuation ring R of a field $\mathbb{K}$ is a subring of $\mathbb{K}$ it is necessarily a domain. We first study valuation rings directly, then show how certain functions that broadly measure some notion of divisibility, called valuations, produce them. Lastly, we compare discrete and non-discrete examples.

Problem 1. Examples of Valuation Rings.
Recall $\mathbb{C}(x, y)$ is the field of formal rational functions in x and y, which concretely is the set of elements f/g where $f, g \in \mathbb{C}[x, y]$ with $g \neq 0$. For each of the following subsets $\mathcal{S} \subset \mathbb{C}(x, y)$ determine whether or not they are valuation rings by showing whether for every $h \in \mathbb{C}(x, y)^\times$ either $h \in \mathcal{S}$ or $h^{-1} \in \mathcal{S}$.

(a) Hint: Use that $\mathbb{C}[x, y]$ is a UFD to understand when g is divisible by y.

$$\mathcal{S}_1 := \left\{ \frac{f}{g} \;\middle|\; f, g \in \mathbb{C}[x, y] \text{ and } g \notin \langle y \rangle \right\}$$

(b) Hint: Consider the element $\frac{x}{y}$.

$$\mathcal{S}_2 := \left\{ \frac{f}{g} \;\middle|\; f, g \in \mathbb{C}[x, y] \text{ and } g(0, 0) \neq 0 \right\}$$

(c) Define the *order* of a non-zero polynomial $f \in \mathbb{C}[x, y]$ to be $\mathbf{ord}(f)$, the lowest degree of a monomial in f and $\mathbf{ord}(0) = \infty$.

$$\mathcal{S}_3 := \left\{ \frac{f}{g} \;\middle|\; f, g \in \mathbb{C}[x, y],\ g \neq 0, \text{ and } \mathbf{ord}(f) \geq \mathbf{ord}(g) \right\}$$

Solution. **TODO:**

(a) yes

(b) no

(c) yes

Problem 2. Ideals in Valuation Rings.
Let R be a valuation ring of $\mathbb{K}$.

(a) Show that for any two non-zero elements $x, y \in R$, either $x|y$ or $y|x$.

(b) Show that every finitely generated ideal of R is principal.

(c) Prove that the ideals of R form a totally ordered set. That is, for any two ideals I, J in R, either $I \subseteq J$ or $J \subseteq I$. (Hint: Given $x \in I$ with $x \notin J$ consider whether an element $y \in J$ can divide x.)

(d) What does the previous parts say about the poset structure on $\mathrm{Spec}(R)$?

Problem 3. Local Rings & Valuation Rings.
An arbitrary ring R is said to be a *local ring* if it has exactly one maximal ideal. We often write a local ring R as a pair $(R, \mathfrak{m})$ where $\mathfrak{m} \subset R$ is the unique maximal ideal.

(a) Describe $\mathrm{mSpec}(R)$ if R is a local ring.

(b) Describe the closed points of $\mathrm{Spec}(R)$ if R is a local ring.

(c) Prove that a valuation ring is a local ring.

Solution.

(a) $\mathrm{mSpec}(R) = \{\mathfrak{m}\}$

(b) the only closed point is the maximal ideal.

(c)

Problem 4. p-Adic Integers.
Fix a prime p. The p-*adic numbers* $\mathbb{Q}_p$ are the completion of $\mathbb{Q}$ with respect to the p-adic absolute value $|\cdot|_p$, defined as follows: for a nonzero rational number x, write

$x = p^k \frac{a}{b}$ where $a, b \in \mathbb{Z}$ are not divisible by p. Then

$$|x|_p = p^{-k}, \quad |0|_p = 0.$$

Note $|\cdot|_p$ extends uniquely to $\mathbb{Q}_p$. The p-*adic integers* are defined by

$$\mathbb{Z}_p = \left\{x \in \mathbb{Q}_p \;\middle|\; |x|_p \leq 1\right\}.$$

Similar to decimal expansion in $\mathbb{R}$, it is a fact that every element $x \in \mathbb{Q}_p$ can be written uniquely as a formal series

$$x = \sum_{i=k}^{\infty} a_i p^i \qquad \text{where} \qquad a_i \in \{0, 1, \dots, p-1\}, \quad a_k \neq 0,$$

and that

$$\mathbb{Z}_p = \left\{\sum_{i=0}^{\infty} a_i p^i \;\middle|\; a_i \in \{0, 1, \dots, p-1\}\right\}.$$

(a) Prove that the p-adic absolute value is well-defined. (Hint: Use the fact that $\mathbb{Z}$ is a UFD to show that every element $x \in \mathbb{Q}$ can be written uniquely (up to sign and reordering) as $p_1^{a_1} \dots p_t^{a_t}$ where the p_i are distinct primes and $a_i \in \mathbb{Z}$.)

(b) Prove the p-adic absolute value satisfies the *non-Archimedean (ultrametric) inequality*:
$$|x + y|_p \leq \max\{|x|_p, |y|_p\}.$$

(c) Prove the p-adic absolute value is multiplicative: $|xy|_p = |x|_p |y|_p$.

(d) Use parts 2b and 3c to show that $\mathbb{Z}_p$ is closed under addition and multiplication.

(e) Conclude that $\mathbb{Z}_p$ is a ring (called the ring of p-adic integers).

(f) Prove that $\mathbb{Z}_p$ is a valuation ring.

The previous exercises suggest why valuation rings are called *valuation* rings: they arise from functions that measure divisibility, generalizing the p-adic absolute value. Such functions are called valuations. An *ordered abelian group* is an abelian group $(\Gamma, +)$ together with a total order $<$ on Γ that is compatible with addition, in the sense that if $\alpha < \beta$ then $\alpha + \gamma < \beta + \gamma$ for all $\alpha, \beta, \gamma \in \Gamma$. We will frequently extend the total ordering on an ordered abelian group Γ to $\Gamma \cup \{\infty\}$ by making the convention that $\gamma < \infty$ for all $\gamma \in \Gamma$. Similarly, we will extend addition to $\Gamma \cup \{\infty\}$ by declaring $\infty + \gamma = \gamma + \infty = \infty$ for all $\gamma \in \Gamma$.

Definition 0.2. *Let* $(\Gamma, +, <)$ *be an ordered abelian group. A* Γ-valued valuation*, or just a* valuation, *on* $\mathbb{K}$ *is a function:* $\nu : \mathbb{K} \to \Gamma \cup \{\infty\}$, *which satisfies the following axioms for all* $x, y \in \mathbb{K}$*:*

1. $\nu(xy) = \nu(x) + \nu(y)$
2. $\nu(x+y) \geq \min\{\nu(x), \nu(y)\}$, *and*
3. $\nu(x) = \infty$ *if and only if* $x = 0$.

Note that this means ν defines a group homomorphism $\nu : \mathbb{K}^\times \to \Gamma$. Indeed, it is standard to define a valuation as a group homomorphism $\nu : \mathbb{K}^\times \to \Gamma$ satisfying axiom (ii), and then extend it to all of $\mathbb{K}$ by setting $\nu(0) = \infty$. This convention is implicit throughout the remainder of the worksheet.

The *value group* of a valuation $\nu : \mathbb{K}^\times \to \Gamma$ is the image of ν. A valuation is a *discrete valuation* if its value group is isomorphic to $\mathbb{Z}$. If $\mathcal{F} \subset \mathbb{K}$ is a subfield, we say that ν is a $\mathcal{F}$-valuation if $\nu(\lambda) = 0$ for all $\lambda \in \mathcal{F}^\times$.

Problem 5. **Examples of Valuations.**
Prove that the following are valuations.

(a) The p-adic valuation:

$$\nu_p : \mathbb{Q}^\times \to \mathbb{Z}, \quad \nu_p\left(p^k \frac{a}{b}\right) = k$$

where $a, b \in \mathbb{Z}$ are not divisible by p.

(b) The order valuation on $\mathbb{C}(x)$:

$$\nu_0 : \mathbb{C}(x)^\times \to \mathbb{Z}, \quad \nu_0(f/g) = \mathbf{ord}_{x=0}(f) - \mathbf{ord}_{x=0}(g).$$

(c) More generally, if $a \in \mathbb{C}$, define

$$\nu_a : \mathbb{C}(x)^\times \to \mathbb{Z}$$

by order of vanishing at $x = a$. Show this is again a valuation.

(d) On $\mathbb{C}(x, y)$, define

$$\nu(f/g) = \mathrm{ord}(f) - \mathrm{ord}(g),$$

where $\mathrm{ord}(h)$ is the degree of the lowest-degree nonzero term of $h \in \mathbb{C}[x, y]$. Show that this is a valuation with value group $\mathbb{Z}$.

Problem 6. Prove that every ordered abelian group is torsion-free. (Hint: Reduce to the case when $x > 0$.)

Problem 7. **Basic Properties of Valuations.**
Let $\nu : \mathbb{K}^\times \to \Gamma$ be an arbitrary valuation. Fix $f, g \in \mathbb{K}^\times$.

(a) Show that $\nu(f^{-1}) = -\nu(f)$.

(b) Show that if $\nu(g) < \nu(f)$, then $\nu(f+g) = \nu(g)$. (Hint: Use that $(f+g)+(-f) = g$.)

(c) Show that $\nu(f) = \nu(-f)$. (Hint: Use part 2b.)

Problem 8. Consider the field $\mathbb{C}(x, y)$. Throughout this question π refers to the number $\pi = 3.1415926...$.

(a) Show that there is a unique $\mathbb{C}$-valuation $\nu_\pi : \mathbb{C}(x, y)^\times \to (\mathbb{R}, +)$ such that

$$\nu_\pi\left(x^a y^b\right) = a + \pi b.$$

What is the value group of ν_π? (Hint: Use Question 3.)

(b) Consider the group $\Gamma = \mathbb{Z} \oplus \mathbb{Z}$ ordered by the lexicographical ordering. Show that there exists a unique $\mathbb{C}$-valuation $\nu : \mathbb{C}(x, y)^\times \to \Gamma$ such that $\nu(x^a y^b) = (a, b)$.

(c) Are either of these valuations constructed in parts 1a or 2b discrete?

Problem 9. **Valuations Determine Valuation Rings.**
Let ν be an arbitrary valuation on $\mathbb{K}$.

(a) Prove that the set $R \subset \mathbb{K}$ of elements $f \in \mathbb{K}$ with $\nu(f) \geq 0$ is a valuation ring of $\mathbb{K}$. We call this the *valuation ring* of ν. (Hint: Remember our convention about $\nu(0)$.)

(b) Show that the following is an ideal of R:

$$\mathfrak{m} \coloneqq \left\{ f \in \mathbb{K} \;\middle|\; 0 < \nu(f) \right\}.$$

(c) Prove that if $f \in R$ and $f \notin \mathfrak{m}$ then f is a unit.

(d) Combine the previous parts to deduce $\mathfrak{m}$ is the unique maximal ideal of R.

(e) Fix an element $\gamma \in \Gamma$. Prove the sets below are ideals of R:

$$\left\{ f \in R \;\middle|\; \nu(f) > \gamma \right\} \quad \text{and} \quad \left\{ f \in R \;\middle|\; \nu(f) \geq \gamma \right\}.$$

Problem 10. Discrete Valuation Rings.
Let $\nu : \mathbb{K}^\times \to \mathbb{Z}$ be a discrete valuation on $\mathbb{K}$. Let $(R, \mathfrak{m})$ be the corresponding *discrete valuation ring.*

(a) Let $I \subset R$ be any non-zero ideal, explain why we can choose an element $f \in I$ such that $\nu(f)$ is minimal among elements in I.

(b) Prove that every ideal in R is principal. In particular, a discrete valuation ring is a PID.

(c) By part 2b there exists an element $t \in R$ such that $\langle t \rangle = \mathfrak{m}$. Show that every ideal of R is either 0 or of the form $\langle t^n \rangle$ for some $n \geq 0$.

Problem 11. Consider the valuation ν_π on $\mathbb{C}(x, y)$ constructed in Question 1a. Let $(R, \mathfrak{m})$ be the corresponding valuation ring for ν_π.

(a) For a real number $x \in \mathbb{R}_{\geq 0}$ let $\{x\} \coloneqq x - \lfloor x \rfloor$ denote its fractional part. Fix a positive integer N and consider the sequence consisting of the fractional parts of the first $N + 1$ multiples of π: $\{0\}, \{\pi\}, \{2\pi\}, \dots, \{N\pi\} \in [0, 1)$. Divide the interval $[0, 1)$ into N evenly spaced subintervals:

$$[0,1) = \left[0, \frac{1}{N}\right) \cup \left[\frac{1}{N}, \frac{2}{N}\right) \cup \cdots \cup \left[\frac{N-1}{N}, 1\right).$$

Prove there exist integers $0 \leq k, j \leq N$ such that $\{k\pi\}$ and $\{j\pi\}$ lie in the same subinterval. Deduce that for any positive integer N there exist integers $0 \leq k, j \leq N$ such that $|\{k\pi\} - \{j\pi\}| \leq \frac{1}{N}$. (Hint: Think about pigeons.)

(b) Using that π is irrational show that $\{j\pi\} \neq \{k\pi\}$ whenever $j \neq k$.

(c) Prove for every real number $\epsilon > 0$, there exist integers $a, b \in \mathbb{Z}$ such that $0 < a + b\pi < \epsilon$.

(d) Prove that $\mathfrak{m}$ is not finitely generated. (Hint: Use part 3c.)

(e) Prove that $\mathfrak{m}$ is the only non-zero radical ideal of R. (Hint: If $I \subset R$ is a non-zero radical ideal, let $f \in I$ be non-zero. For an arbitrary element $h \in \mathfrak{m}$ show that $h^n \in \langle f \rangle$ for $n \gg 0$.)

(f) Describe Spec(V).

Problem 12. Consider the valuation ν on $\mathbb{C}(x, y)$ constructed in Question 2b. Let $(R, \mathfrak{m})$ be the corresponding valuation ring for ν.

(a) Show that the maximal ideal of R is

$$\mathfrak{m} = \left\{ f \in \mathbb{C}(x, y) \;\middle|\; \nu(f) \geq (0, 1) \right\} = \langle y \rangle.$$

(b) Show that the ideal $\langle x \rangle \subset R$ is not prime. (Hint: $x = xy^{-1} \cdot y$.)

(c) Show that the set below is a prime, non-maximal ideal of R:

$$P := \left\{ f \in R \;\middle|\; \nu(f) = (a, b) \text{ where } a \geq 1 \right\} \cup \{0\}.$$

(d) Use the valuation to construct an infinite strictly ascending chain of ideals in R:

$$\left\langle xy^{-1} \right\rangle \subset \left\langle xy^{-2} \right\rangle \subset \left\langle xy^{-3} \right\rangle \subset \cdots .$$

Explain why all inclusions are strict.

Problem 13. **Valuation Rings Determine Valuations.**
Let R be a valuation ring of $\mathbb{K}$ with $R \neq \mathbb{K}$. This problem shows how to recover a valuation from the divisibility structure of R.

(a) Let Γ' be the set of nonzero principal ideals of R:

$$\Gamma' = \left\{ \langle a \rangle \subset R \;\middle|\; a \in R,\ a \neq 0 \right\}.$$

Define a multiplication on Γ' by ideal multiplication: $\langle a \rangle \cdot \langle b \rangle = \langle ab \rangle$.

  (a) Show that this operation is well-defined and makes Γ' into an abelian monoid (associative, commutative, binary operation with an identity element).

  (b) Show that Γ' is totally ordered by inclusion: for distinct elements $\langle a \rangle, \langle b \rangle \in \Gamma'$, either $\langle a \rangle \subseteq \langle b \rangle$ or $\langle b \rangle \subseteq \langle a \rangle$.

(b) Define an equivalence relation $\sim$ on $\mathbb{K}^\times$ by $x \sim y$ if and only if $x = uy$ for some unit $u \in R^\times$. Let $\Gamma = \mathbb{K}^\times / \sim$ denote the set of equivalence classes. Given $x \in \mathbb{K}^\times$ write $[x]$ its equivalence class in Γ.

  (i) Show that $\sim$ is a well-defined equivalence relation.

  (ii) Show that multiplication in $\mathbb{K}^\times$ induces a well-defined operation on Γ, making it into an abelian group.

(iii) Define $[x] \leq [y]$ if and only if $yx^{-1} \in R$. Show this is a well-defined total order on Γ.

(iv) Deduce that Γ is an ordered abelian group.

(v) Show that the map $\Gamma' \to \Gamma$ given by $\langle a \rangle \mapsto [a]$ is an order-preserving embedding whose image consists exactly of the elements $\geq [1]$.

(c) Define $v : \mathbb{K}^\times \to \Gamma$ by $v(x) = [x]$. Prove that v is a valuation on $\mathbb{K}$ with value group Γ and

$$R = \left\{ x \in \mathbb{K} \;\middle|\; v(x) \geq [1] \right\}.$$

Moreover, $v(x) \in \Gamma'$ if and only if $x \in R$.